

Session 3: Internet & Networking Basics Internet Fundamentals: How computers communicate. Protocols: Define HTTP, HTTPS, FTP, SMTP (overview only, 5 mins each). Domain & DNS: Explain how domain names map to IP addresses. Demonstrate using nslookup google.com. Explain DNS hierarchy: root, TLD, domain, subdomain. IP Addressing: What is an IP (IPv4 vs IPv6)? HTTP Lifecycle: Request? Response? Status Codes (focus on 200, 404, 500). Trainer Demo: Use browser DevTools? Network tab

TASKS:

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

HTTP: HyperText Transfer Protocol

Used to transfer web pages between a web browser and a web server.

HTTPS: HyperText Transfer Protocol Secure

Used to securely transfer web pages and protect data over the internet.

FTP: File Transfer Protocol

Used to upload and download files between computers over a network.

SMTP: Simple Mail Transfer Protocol

Used to send emails from one email server to another.

2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

IPv4:

Uses a 32-bit address (e.g., 192.168.1.1).

Has a limited number of IP addresses.

Supports manual configuration and DHCP.

IPv6:

Uses a 128-bit address (e.g., 2001:db8::1).

Provides a much larger number of IP addresses.

Supports Automatic (SLAAC).

3. Run nslookup google.com on your system and write down the IP address you get.

IPv6 Addresses:

2404:6800:4009:807::200e

IPv4 Addresses:

142.250.70.46

4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

200: OK – The request was successful, and the page loaded correctly.

301: Moved Permanently – The page has been permanently moved to a new URL.

404: Not Found – The requested page or resource could not be found.

500: Internal Server Error – The server encountered an error while processing the request.

403: Forbidden – Access to the requested page or resource is denied.

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

Root (.) – The highest level in DNS, represented by a dot (.) at the end of a domain name.

TLD (Top-Level Domain) – The last part of the domain name, such as com.

Domain – The main name registered under the TLD, such as google.

Subdomain – A prefix added before the domain, such as www.

Example: www.google.com

Root: (.)

TLD: com

Domain: google

Subdomain: www